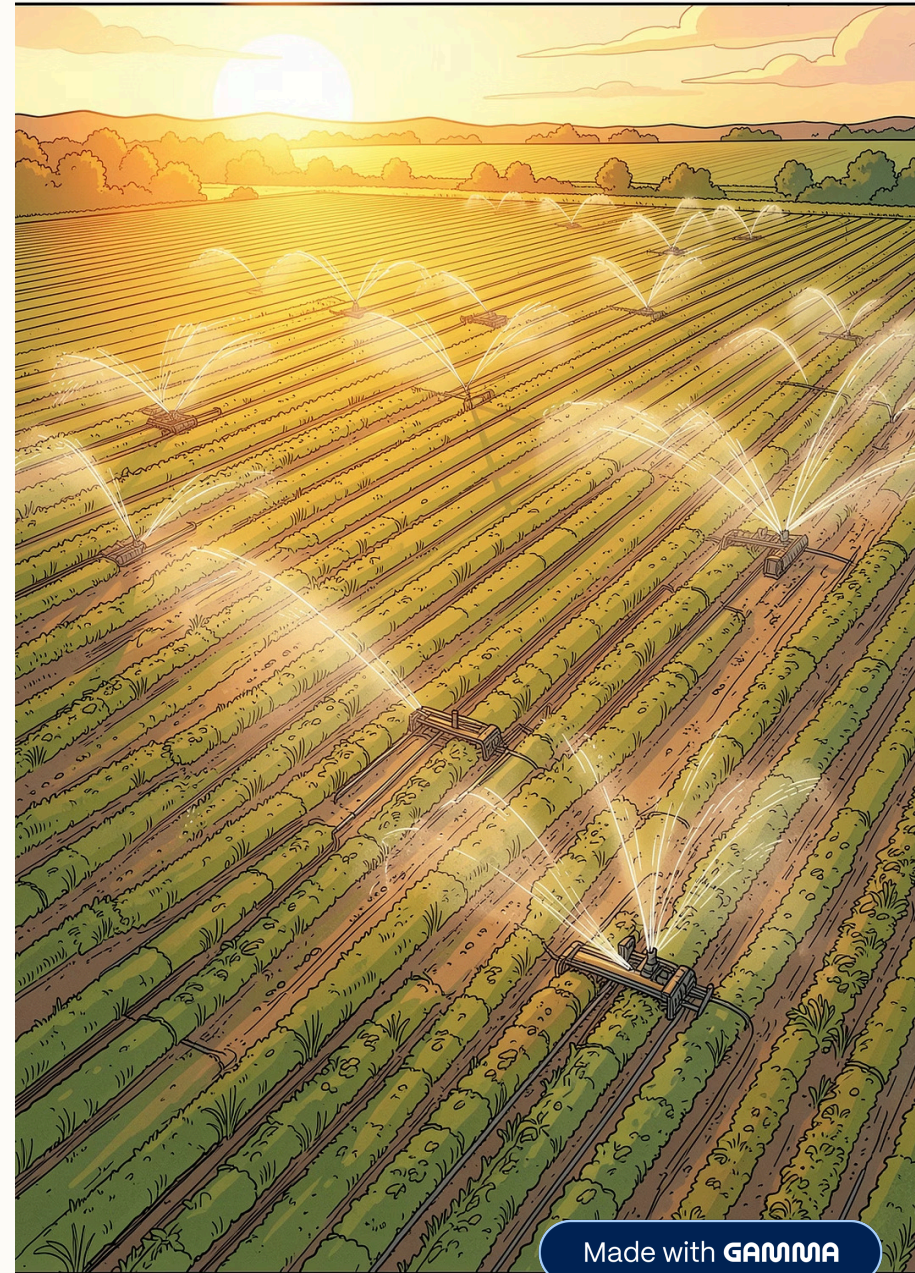
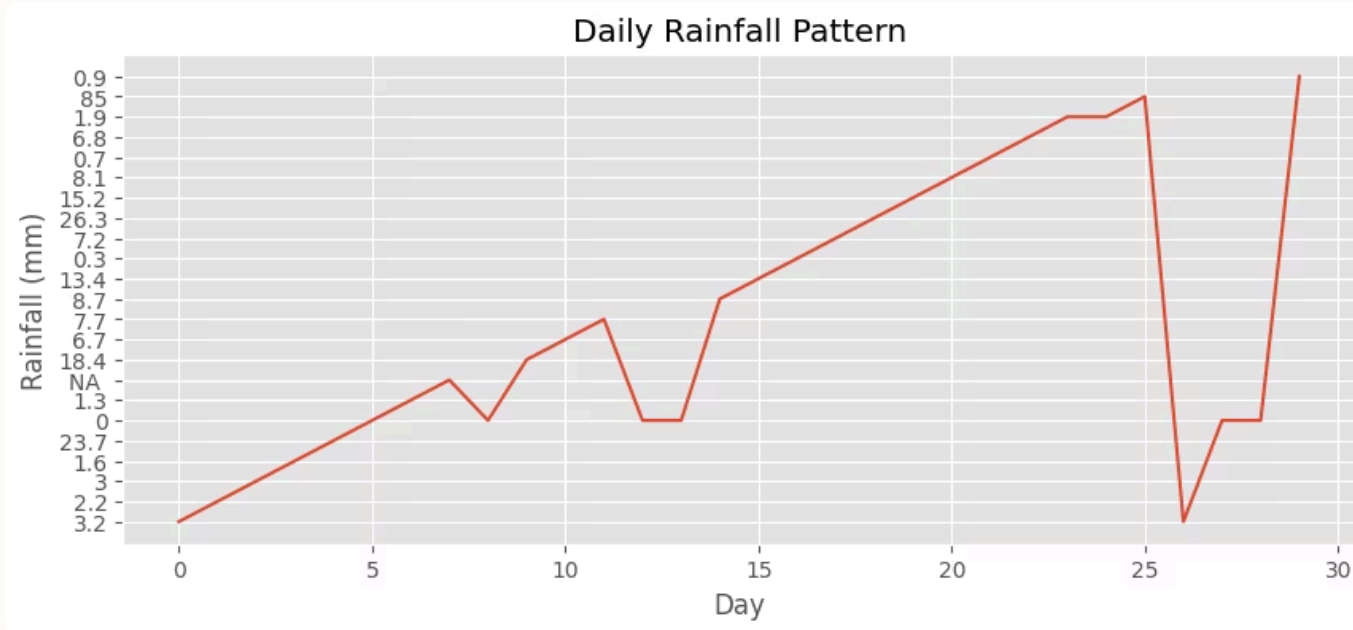


HYDROSENSE KENYA VISUALISATION SLIDES

A data-driven exploration of rainfall trends, temperature variation, soil moisture dynamics, and numerical simulation methods for optimized agricultural water management.

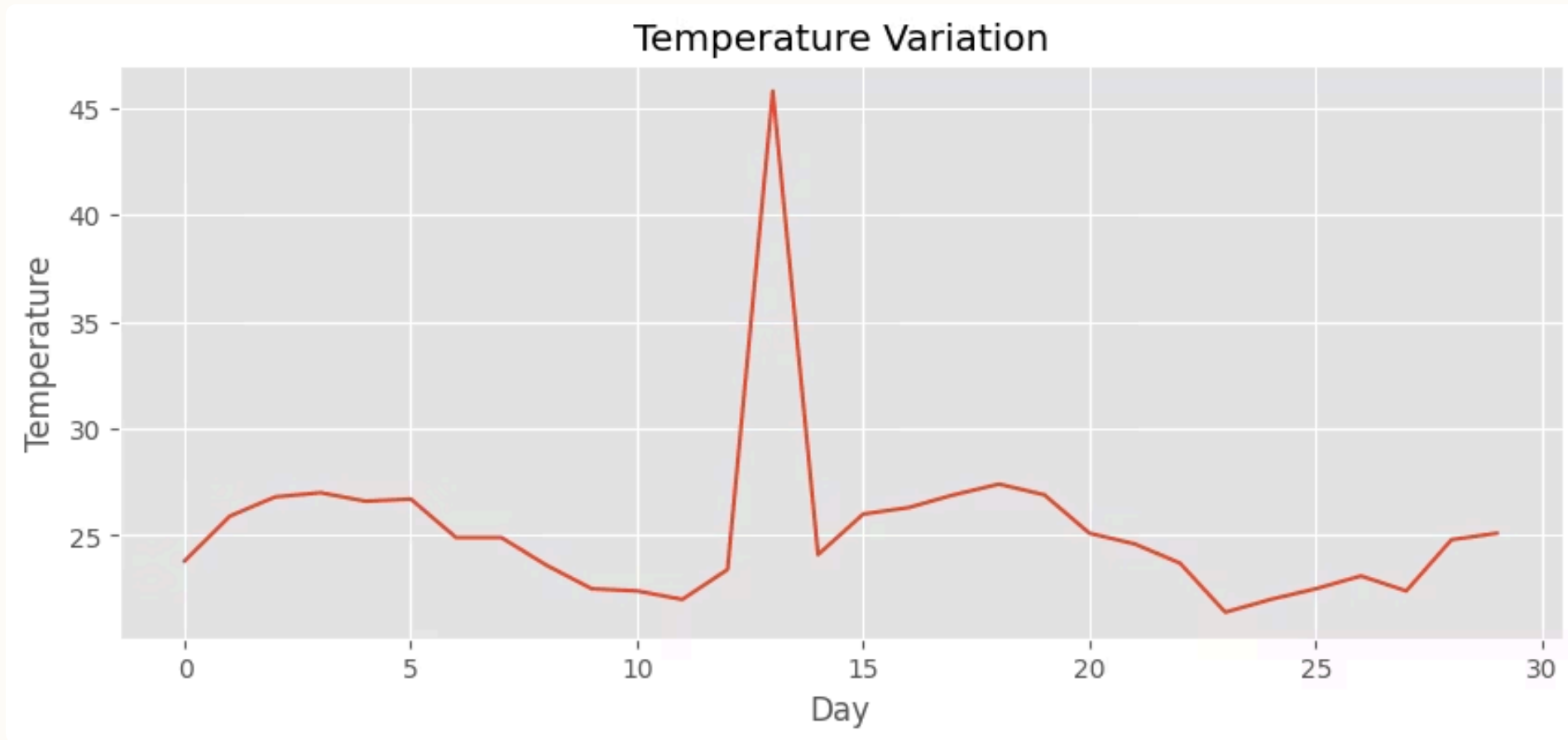


Rainfall Trend



What to Look For

Examine the chart for seasonal rainfall patterns, periods of deficit or surplus, and any long-term trends that may affect irrigation planning. Identifying these patterns helps determine when supplemental irrigation is most critical and where rainfall alone may suffice.



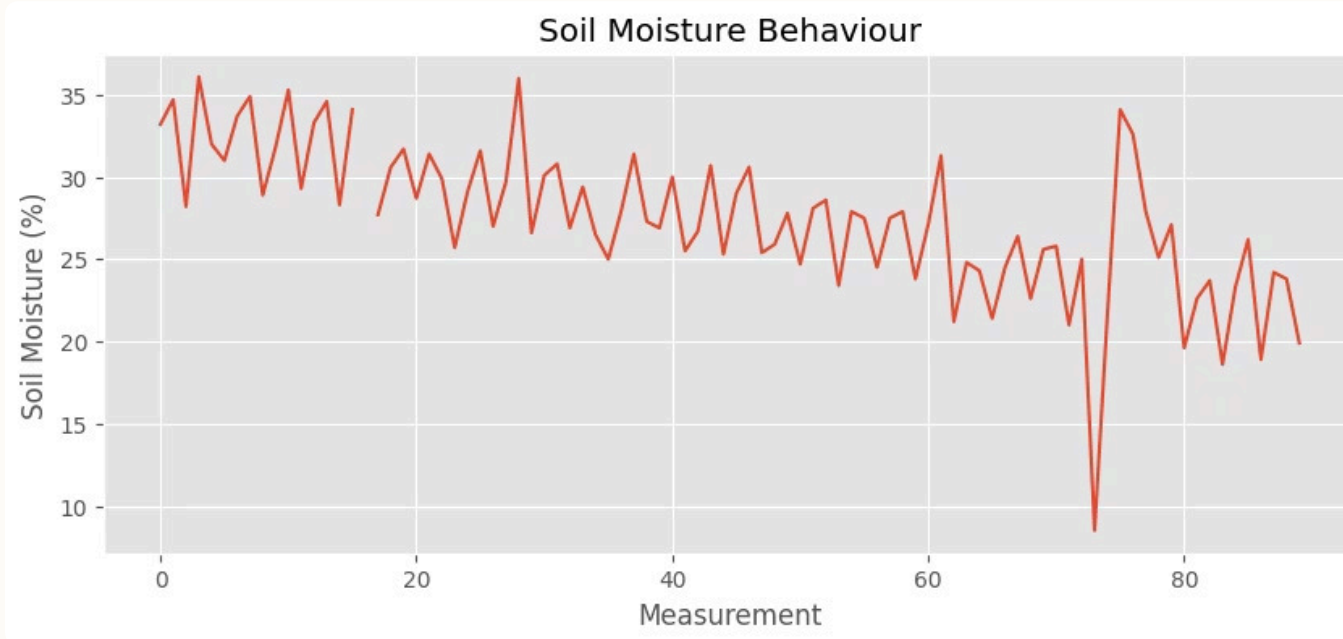
VISUALIZATION 2

Temperature Variation

Temperature directly drives evapotranspiration rates, influencing how quickly soil moisture is depleted. Cross-referencing this chart with rainfall and soil moisture data reveals periods of high water demand and helps refine irrigation timing.

VISUALIZATION 3

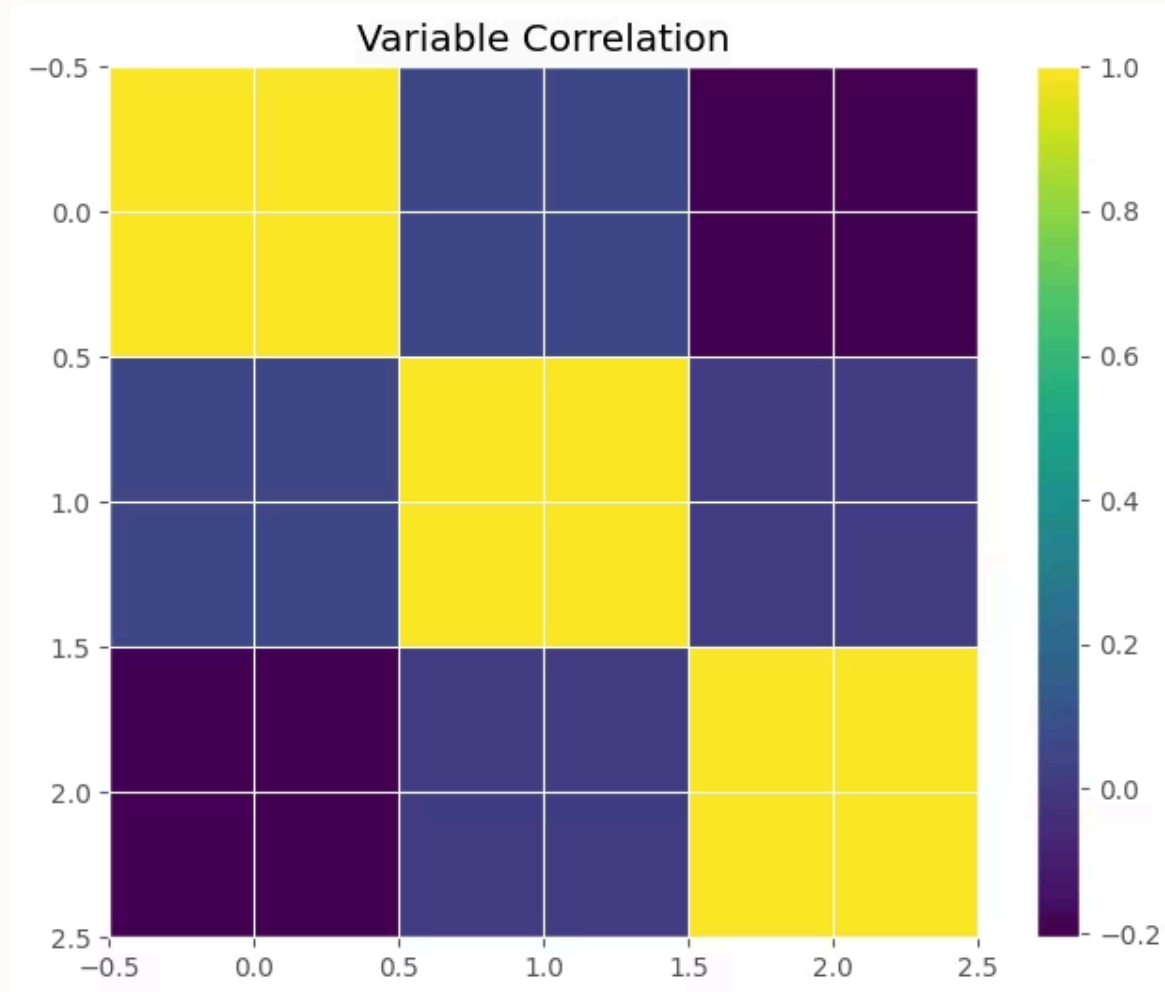
Soil Moisture Trend



Why It Matters

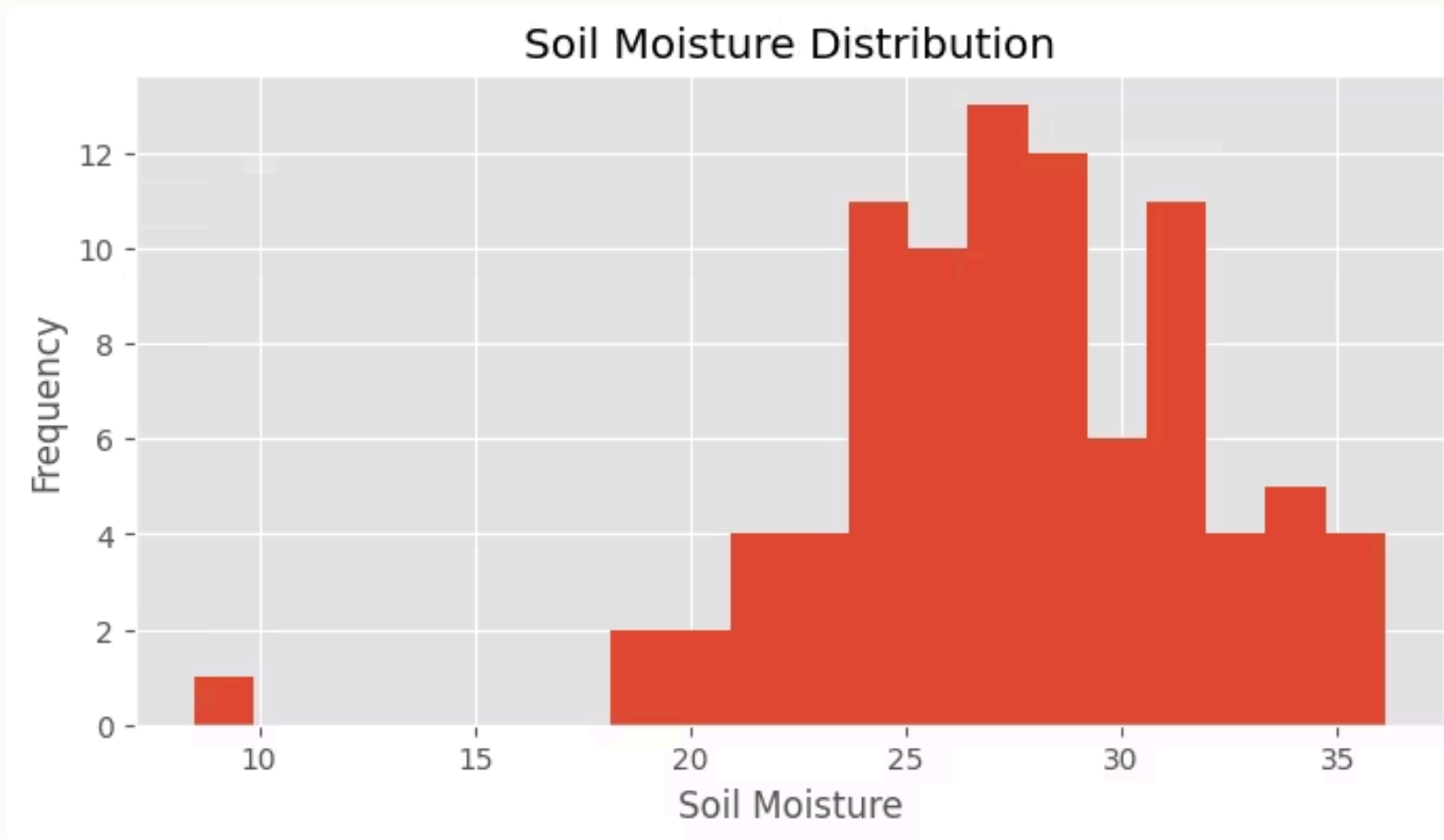
Soil moisture is the central variable in irrigation scheduling. This trend reveals depletion and recharge cycles, showing how rainfall events and temperature peaks translate into actual field conditions. Use it to validate irrigation timing decisions.

Correlation Heatmap



Reading the Heatmap

The heatmap quantifies relationships between rainfall, temperature, and soil moisture. Strong correlations confirm which drivers most directly impact soil moisture levels, while weak correlations may indicate lag effects or confounding variables worth investigating further.

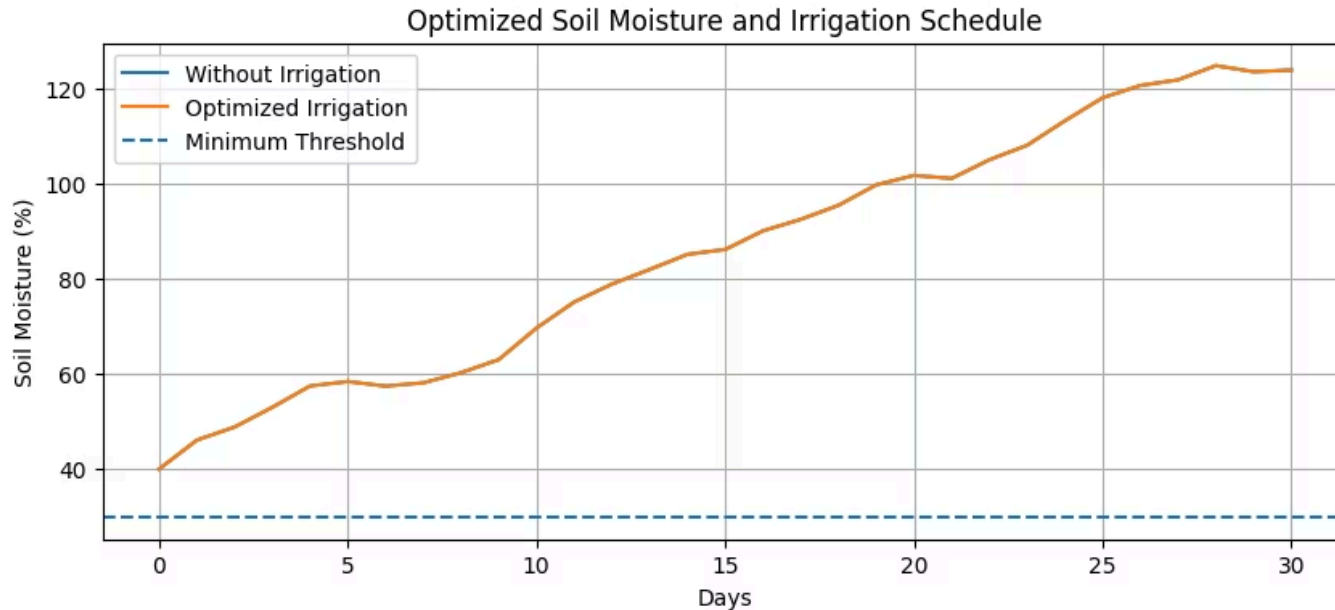


VISUALIZATION 5

Soil Moisture Distribution

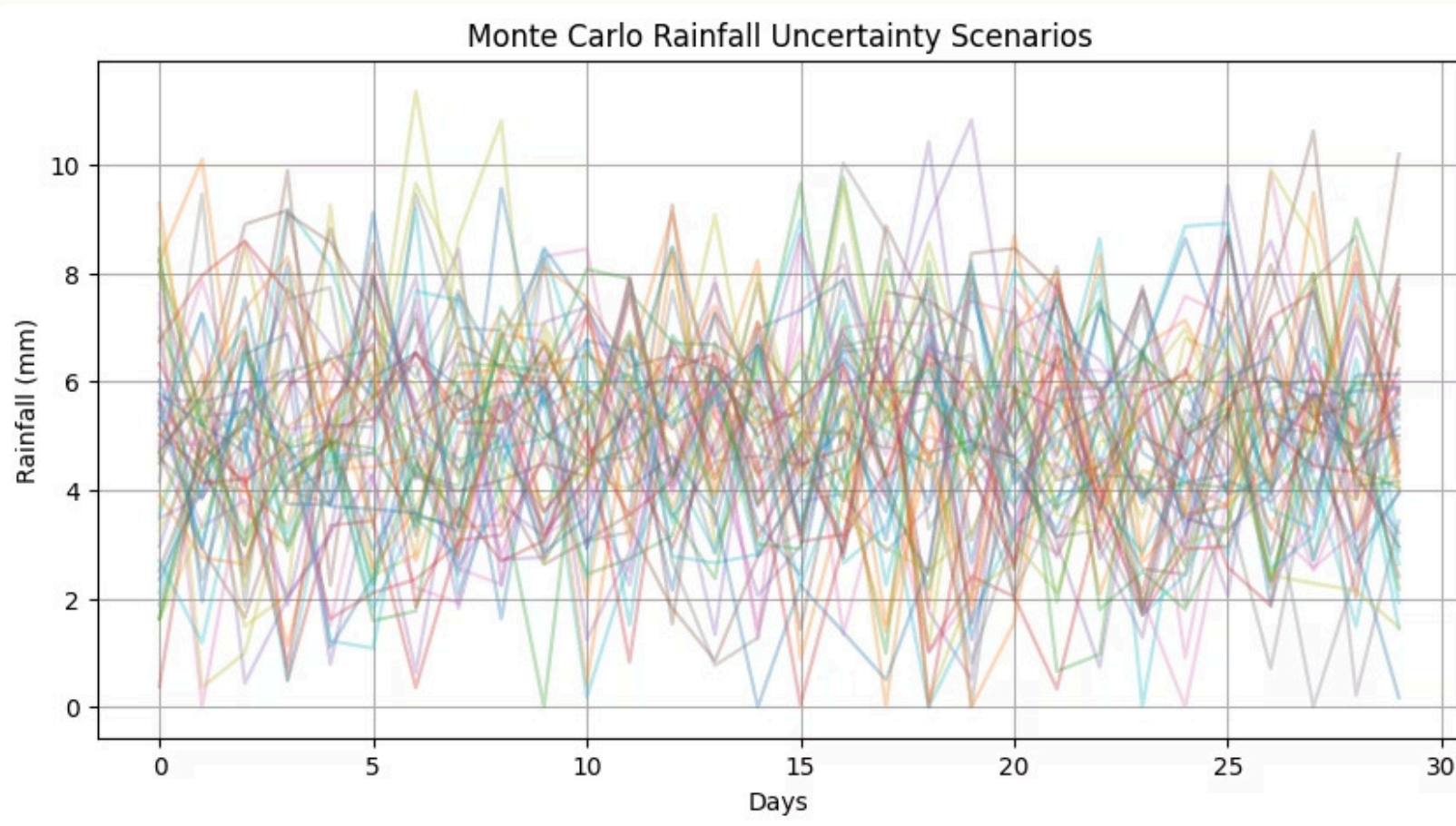
The distribution reveals the spread and variability of soil moisture values across the dataset. Understanding this spread helps assess how consistently fields maintain adequate moisture and identifies the frequency of conditions that may trigger irrigation events.

Optimized Irrigation Schedule



Schedule Insights

The optimized schedule integrates rainfall forecasts, temperature-driven evapotranspiration, and soil moisture thresholds to recommend precise irrigation timing and volumes. This approach minimizes water waste while maintaining crop health.

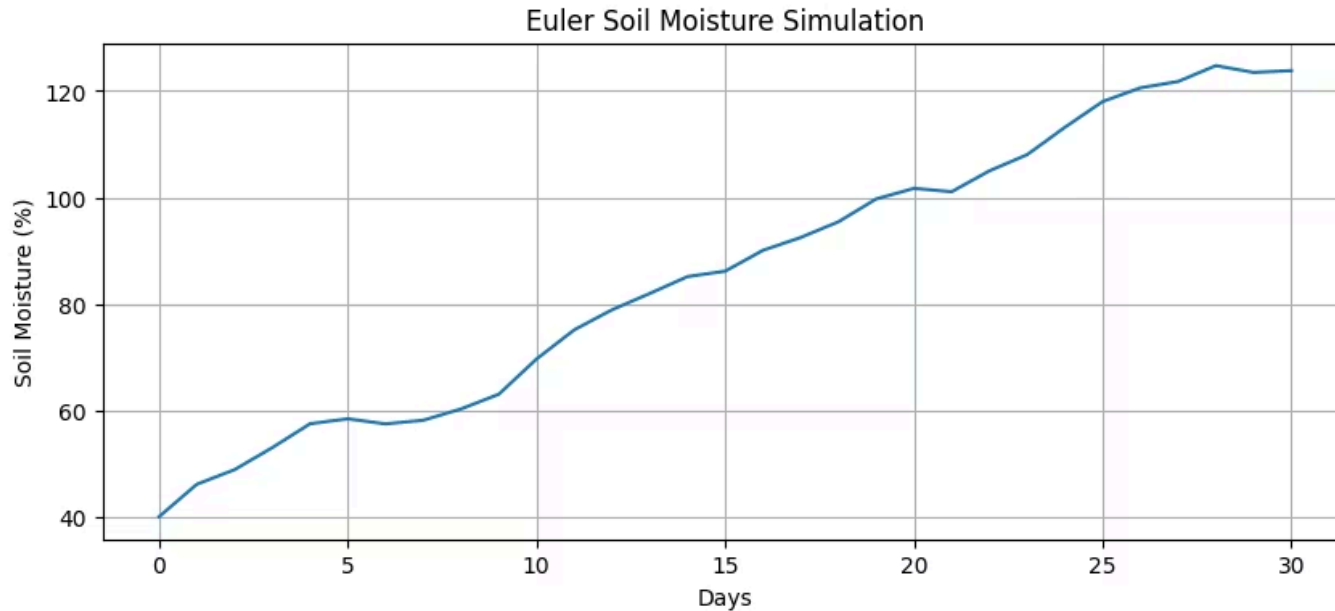


UNCERTAINTY ANALYSIS

Monte Carlo Rainfall Uncertainty

The Monte Carlo analysis quantifies how rainfall variability propagates through the irrigation model. The spread of simulated outcomes indicates the confidence range for scheduling decisions, helping planners account for uncertainty in precipitation forecasts.

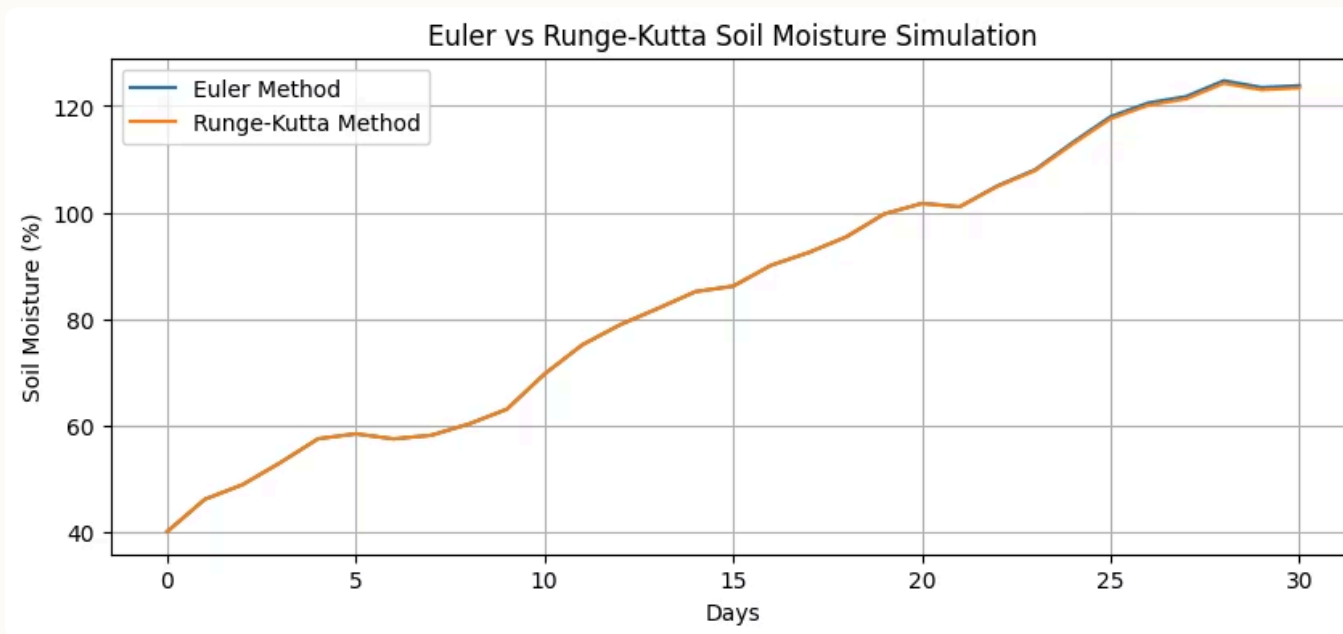
Euler Method Simulation



How It Works

The Euler method approximates soil moisture dynamics by stepping forward in time using the current rate of change. It provides a straightforward numerical solution for modeling moisture depletion and recharge between irrigation events.

Euler vs. RK4 & Numerical Convergence



Key Takeaways

The comparison chart shows how Euler and RK4 methods differ in accuracy for soil moisture simulation. The convergence analysis confirms that reducing step size improves solution accuracy, validating the numerical approach used in the irrigation model.